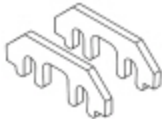


Camshaft — LH



Special Tool(s)

 ST3234-A	Hold Down, Secondary Chain 303-1530
 ST2979-A	Tool, Camshaft Holding 303-1248

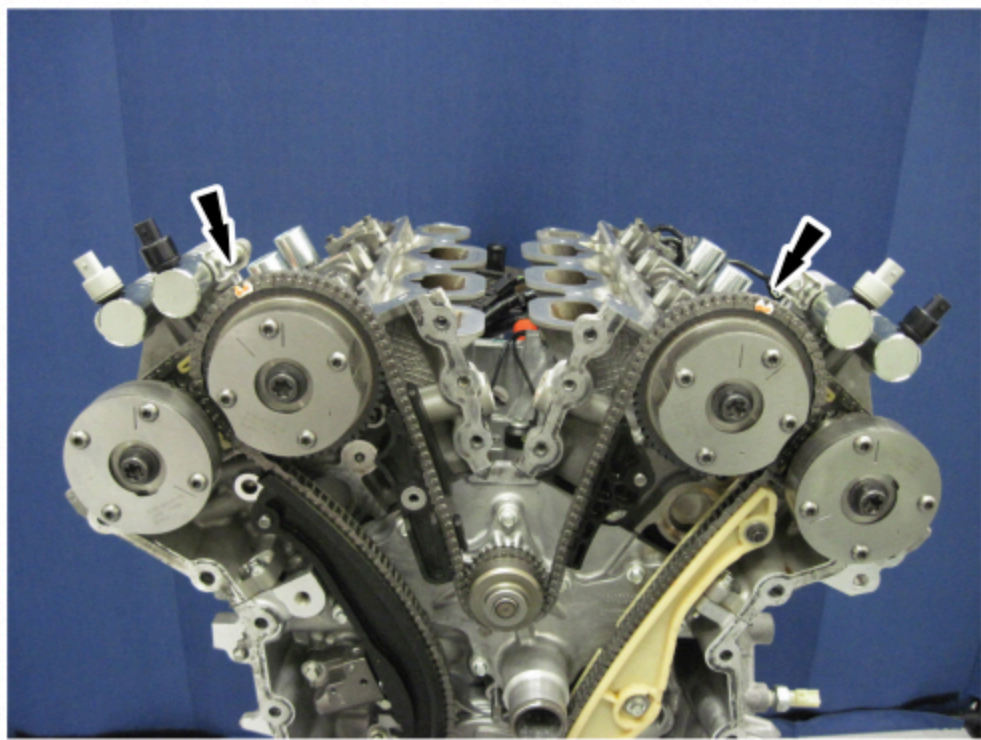
Material

Item	Specification
Motorcraft® SAE 5W-30 Synthetic Blend Motor Oil (US); Motorcraft® SAE 5W-30 Super Premium Motor Oil (Canada) XO-5W30-QSP (US); CXO-5W30-LSP12 (Canada)	WSS-M2C946-A

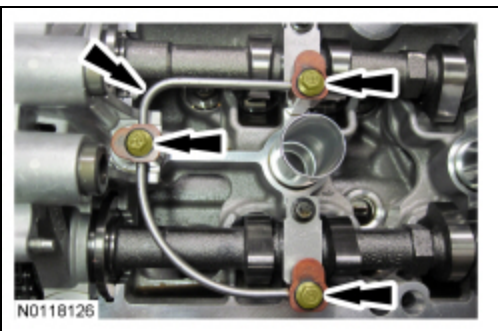
Removal

NOTICE: During engine repair procedures, cleanliness is extremely important. Any foreign material, including any material created while cleaning gasket surfaces that enters the oil passages, coolant passages or the oil pan, may cause engine failure.

1. Remove the engine front cover. For additional information, refer to Engine Front Cover in this section.
2. Rotate the crankshaft clockwise and align the timing marks on the VCT assemblies as shown.

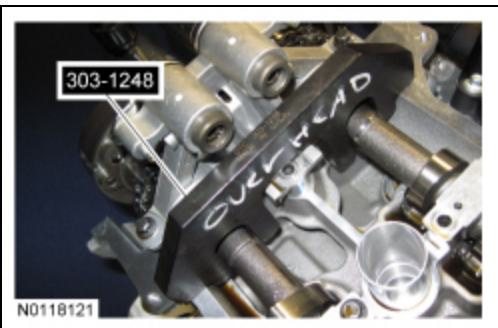


3. Remove the 3 bolts and the LH valve train oil tube.

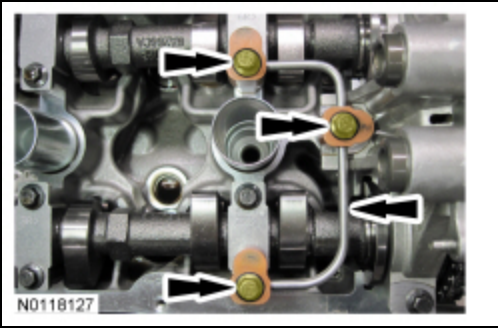


4. **NOTE:** *The Camshaft Holding Tool will hold the camshafts in the Top Dead Center (TDC) position.*

Install the Camshaft Holding Tool onto the flats of the LH camshafts.

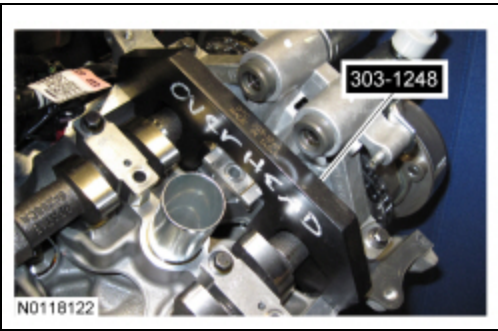


5. Remove the 3 bolts and the RH valve train oil tube.



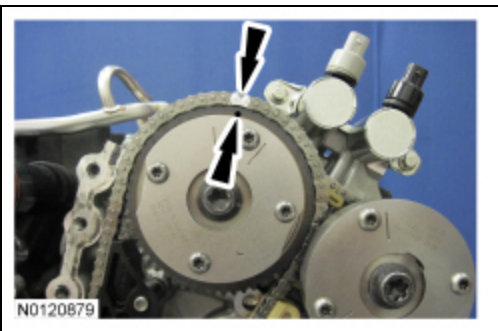
6. **NOTE:** The Camshaft Holding Tool will hold the camshafts in the TDC position.

Install the Camshaft Holding Tool onto the flats of the RH camshafts.

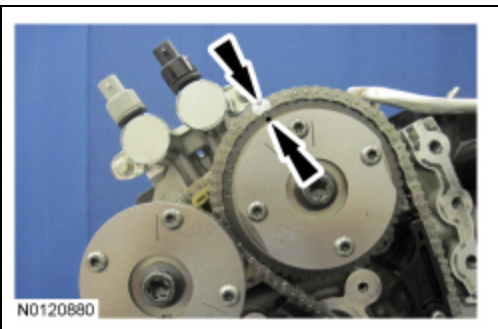


NOTE: The following 3 steps are for primary timing chains that the colored links are not visible.

7. Mark the timing chain link that aligns with the timing mark on the LH intake VCT assembly as shown.

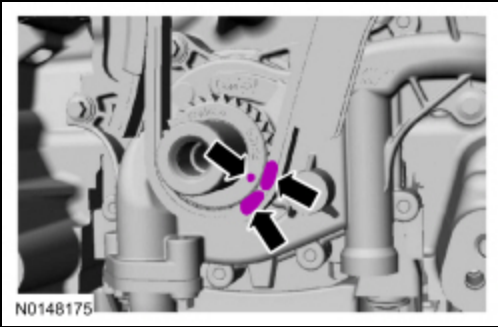


8. Mark the timing chain link that aligns with the timing mark on the RH intake VCT assembly as shown.

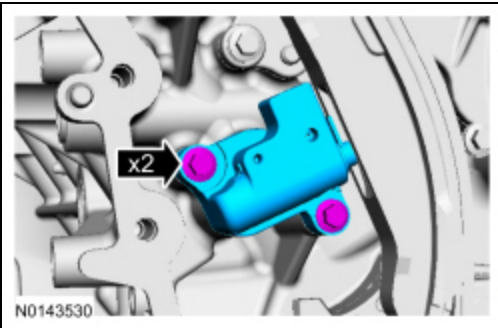


9. **NOTE:** The crankshaft sprocket timing mark should be between the 2 colored links.

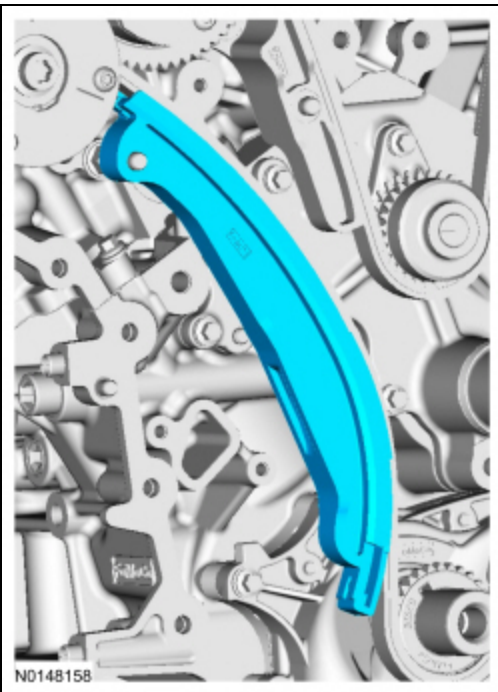
Mark the 2 timing chain links that align with the timing mark on the crankshaft sprocket as shown.



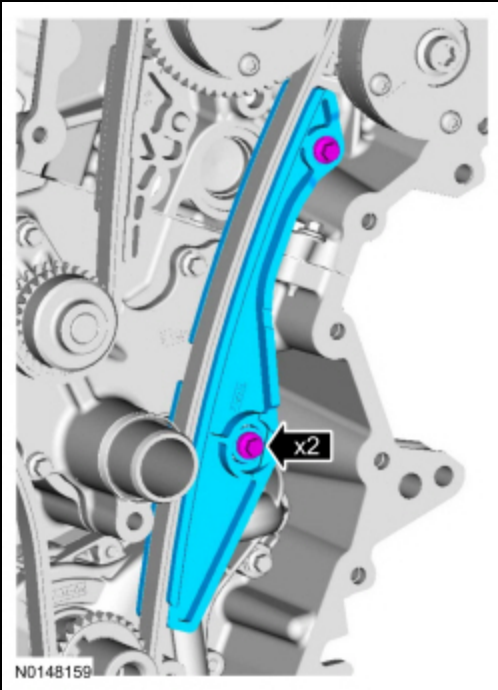
10. Remove the 2 bolts and the primary timing chain tensioner.



11. Remove the primary timing chain tensioner arm.



12. Remove the 2 bolts and the lower LH primary timing chain guide.

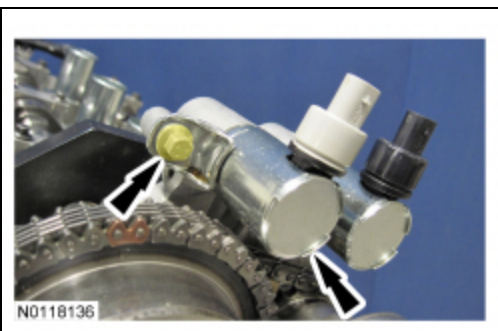


13. **NOTE:** Removal of the VCT oil control solenoid will aid in the removal of the primary timing chain.

NOTE: A slight twisting motion will aid in the removal of the VCT oil control solenoid.

NOTE: Keep the VCT oil control solenoid clean of dirt and debris.

Remove the bolt and the LH intake VCT oil control solenoid.

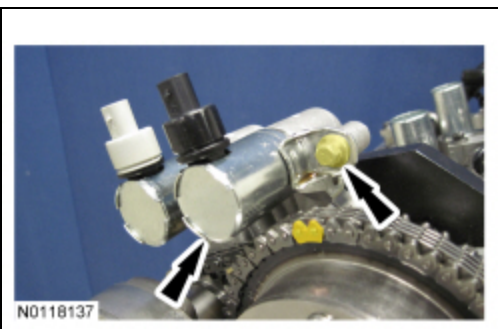


14. **NOTE:** Removal of the VCT oil control solenoid will aid in the removal of the primary timing chain.

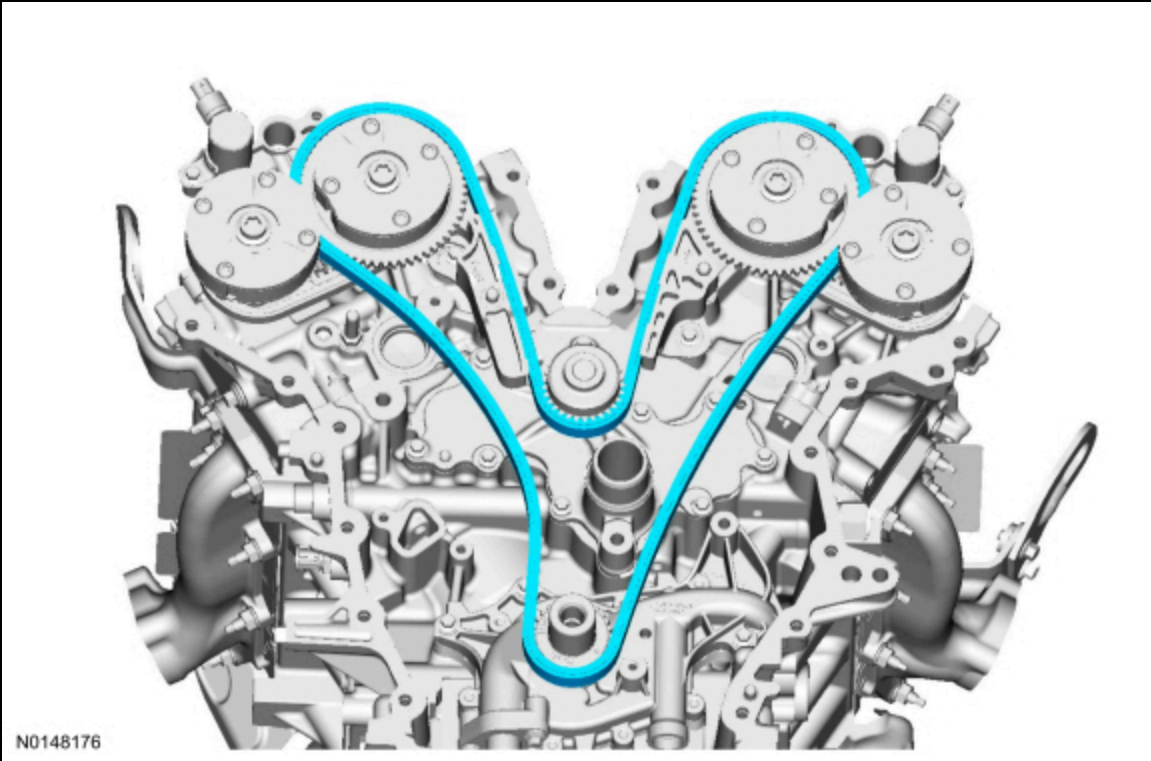
NOTE: A slight twisting motion will aid in the removal of the VCT oil control solenoid.

NOTE: Keep the VCT oil control solenoid clean of dirt and debris.

Remove the bolt and the RH intake VCT oil control solenoid.



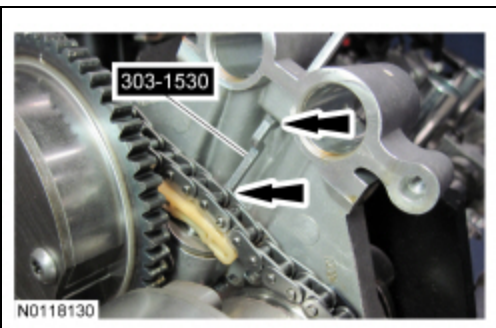
15. Remove the primary timing chain.



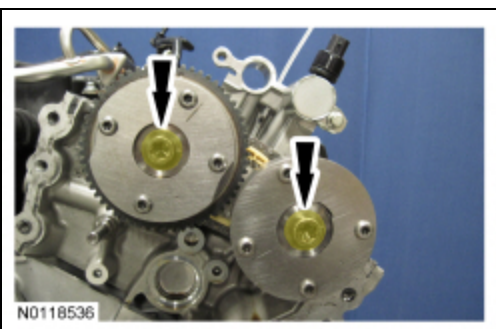
16. **NOTE:** The 2 LH VCT oil control solenoids are removed for clarity.

NOTE: The Secondary Chain Hold Down is inserted through a hole in the top of the mega cap.

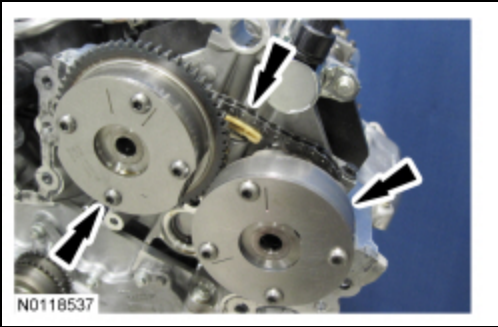
Compress the LH secondary timing chain tensioner and install the Secondary Chain Hold Down in the hole on the rear of the secondary timing chain tensioner guide and let it hold against the mega cap to retain the tensioner in the collapsed position.



17. Remove and discard the 2 LH VCT assembly bolts.

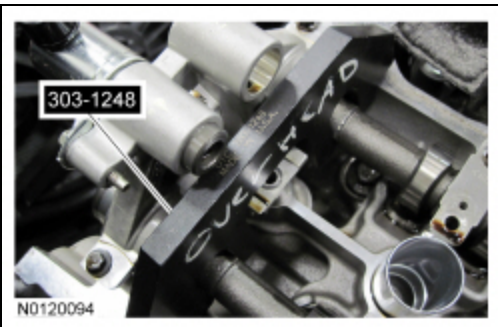


18. Remove the 2 LH VCT assemblies and secondary timing chain.



19. **NOTE:** When the Camshaft Holding Tool is removed, valve spring pressure may rotate the LH camshafts approximately 3 degrees to a neutral position.

Remove the Camshaft Holding Tool from the LH camshafts.

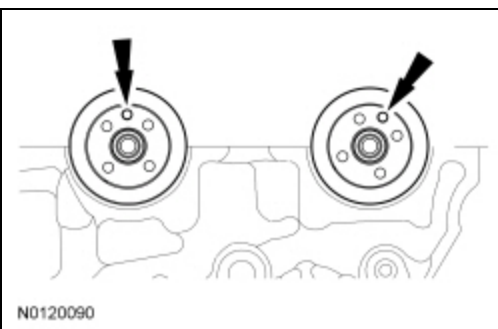


20. Remove the 12 camshaft cap bolts, the 6 camshaft caps, and the mega cap.
- Remove the LH camshafts.

Installation

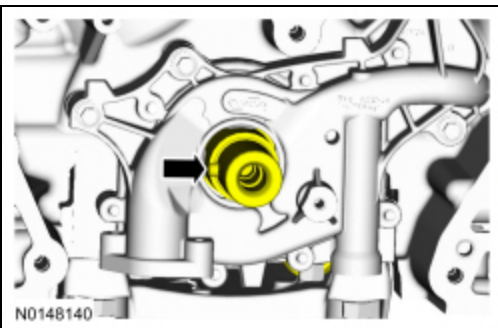
1. **NOTE:** Coat the camshafts with clean engine oil prior to installation.

Position the camshafts onto the LH cylinder head in the neutral position as shown.



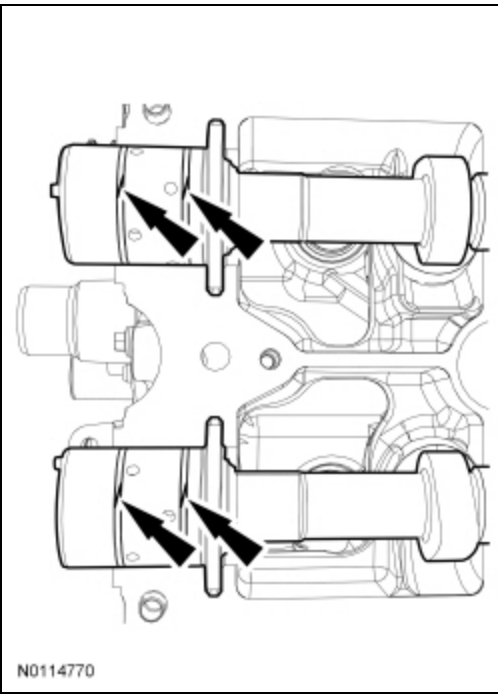
2. **NOTICE:** The crankshaft must remain in the free wheeling position (crankshaft dowel pin at 9 o'clock) until the camshafts are installed and the valve clearance is checked/ adjusted. Do not turn the crankshaft until instructed to do so. Failure to follow this process will result in severe engine damage.

Rotate the crankshaft counterclockwise until the crankshaft dowel pin is in the 9 o'clock position.



3. **NOTICE:** The camshaft seal gaps must be at the 12 o'clock position or damage to the engine may occur.

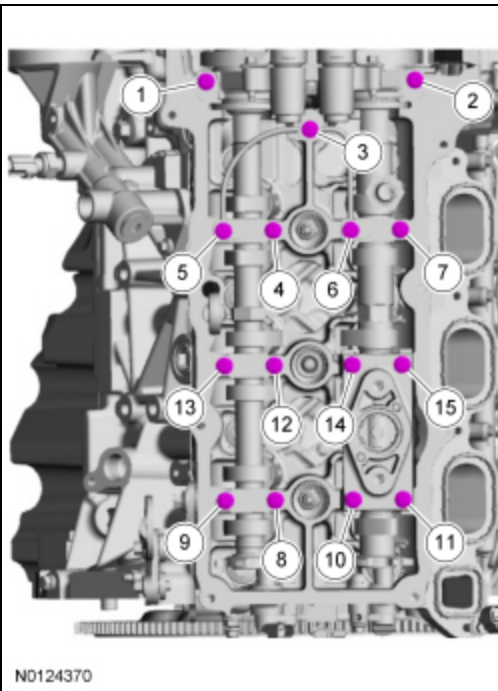
Position the 4 camshaft seals gaps as shown.



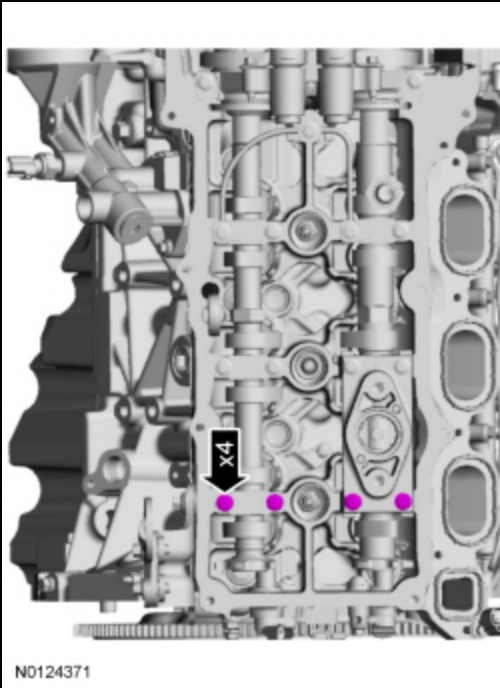
4. **NOTE:** *Cylinder head camshaft bearing caps are numbered to verify that they are assembled in their original positions.*

Install the 5 camshaft caps, mega cap, valve train oil tube and the 15 bolts in the sequence shown.

- Tighten to 8 Nm (71 lb-in) then additional 45 degrees.

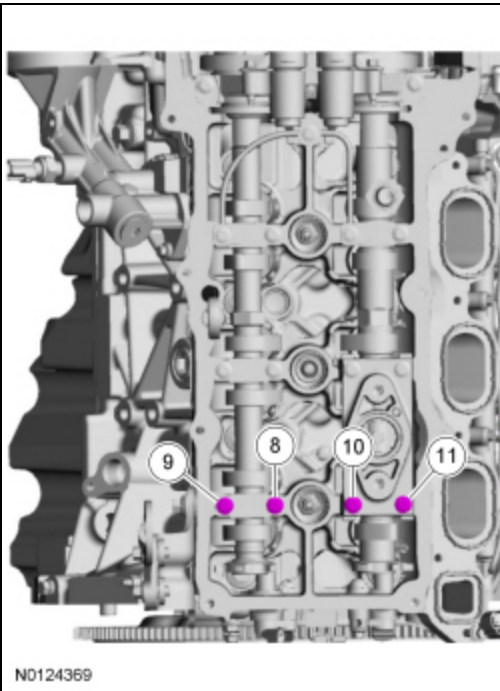


5. Loosen the 4 camshaft caps bolts.



6. Tighten the 4 camshaft caps bolts in the sequence shown.

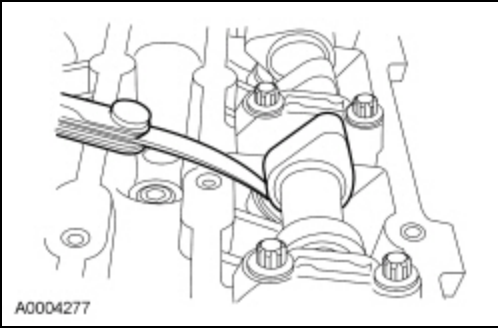
- Tighten bolts 8, 9, 10 and 11 to 8 Nm (71 lb-in) then additional 45 degrees.



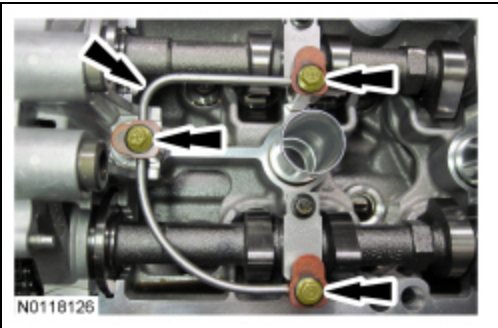
7. **NOTICE:** If any components are installed new, the engine valve clearance must be checked/adjusted or engine damage may occur.

NOTE: Use a camshaft sprocket bolt to turn the camshafts.

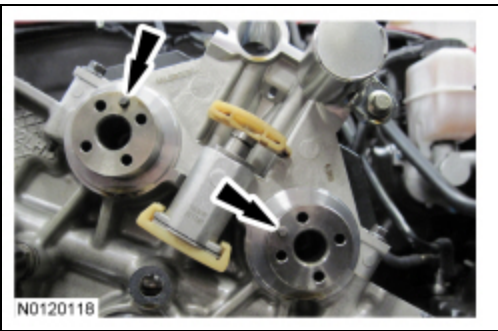
Using a feeler gauge, confirm that the valve tappet clearances are within specification. If valve tappet clearances are not within specification, the clearance must be adjusted by installing new valve tappet(s) of the correct size. For additional information, refer to Valve Clearance Check in this section.



8. Remove the 3 bolts and the LH valve train oil tube.

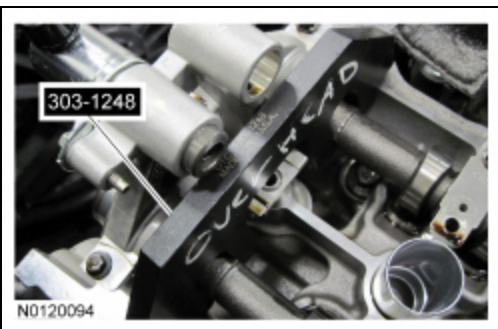


9. Rotate the LH camshafts to the TDC position as shown.

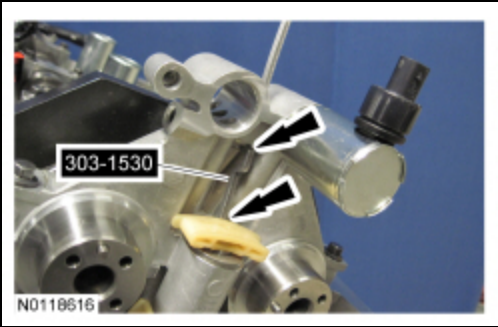


10. **NOTE:** *The Camshaft Holding Tool will hold the camshafts in the TDC position.*

Install the Camshaft Holding Tool onto the flats of the LH camshafts.

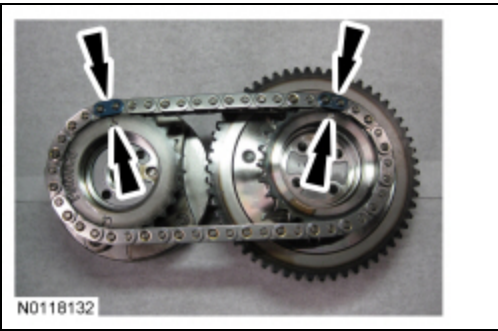


11. Compress the LH secondary timing chain tensioner and install the Secondary Chain Hold Down to retain the tensioner in the collapsed position.



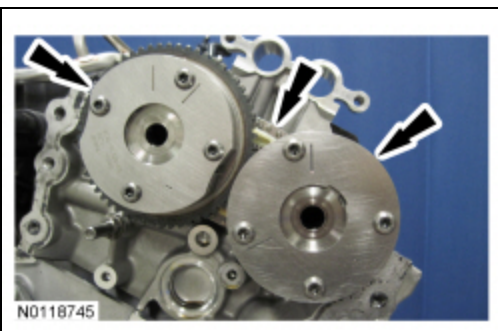
12. Assemble the 2 LH VCT assemblies and the LH secondary timing chain.

- Align the colored links with the timing marks.



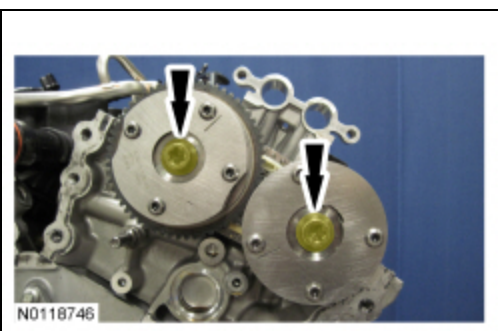
13. **NOTE:** *It may be necessary to rotate the camshafts slightly, to install the LH secondary timing assembly.*

Position the 2 LH VCT assemblies and secondary timing chain onto the camshafts by aligning the holes in the VCT assemblies with the dowel pins in the camshafts.



14. Install the 2 new LH VCT bolts and tighten in 4 stages.

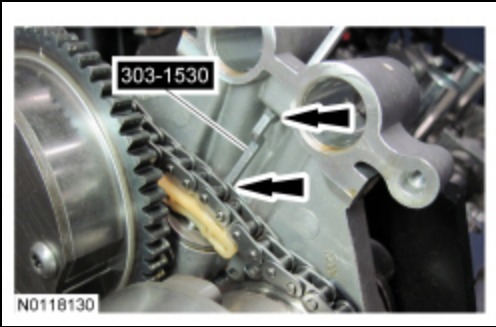
- Stage 1: Tighten to 40 Nm (30 lb-ft).
- Stage 2: Loosen one full turn.
- Stage 3: Tighten to 25 Nm (18 lb-ft).
- Stage 4: Tighten an additional 180 degrees.



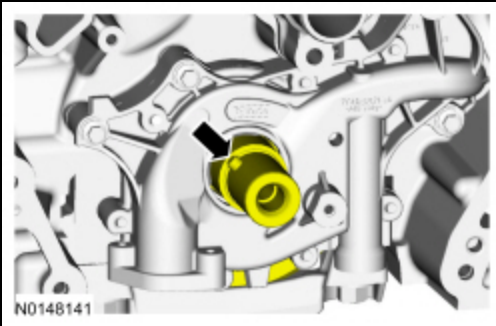
15. **NOTE:** *The 2 VCT oil control solenoids are removed for clarity.*

Compress the LH secondary timing chain tensioner and remove the Secondary Chain Hold Down.

- Make sure the secondary timing chain is centered on the timing chain tensioner guides.



16. Rotate the crankshaft clockwise 60 degrees to **T.D.C.** position (crankshaft dowel pin at the 11 o'clock position).



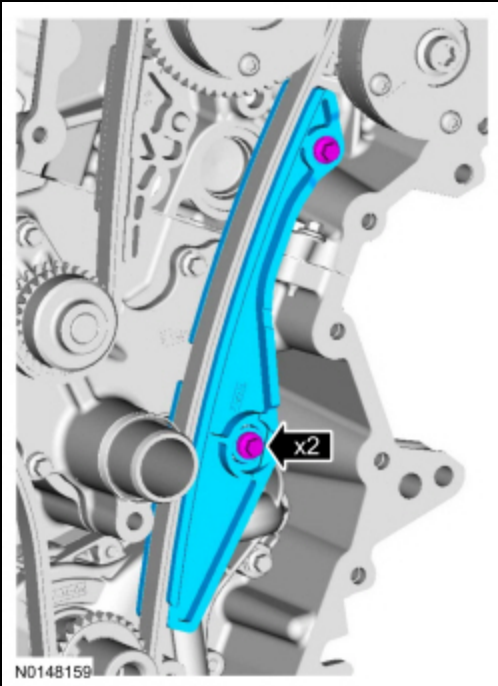
17. **NOTE:** *It may be necessary to rotate the camshafts slightly, to align the timing marks.*

Install the primary timing chain with the colored links aligned with the timing marks on the **V.C.T.** assemblies and the crankshaft sprocket.

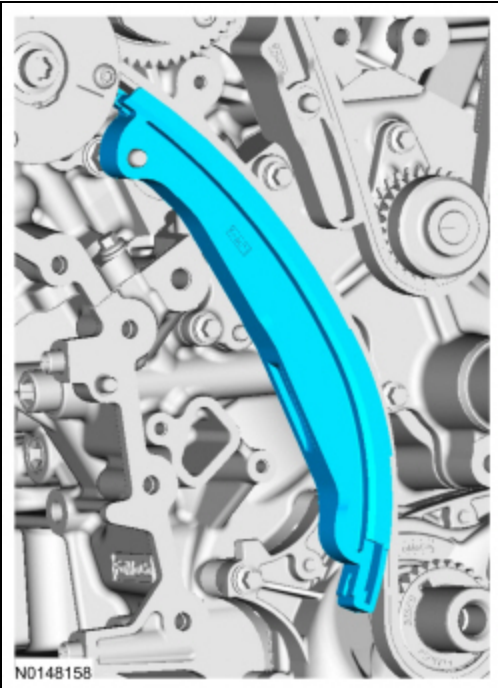


18. Install the lower LH primary timing chain guide and the 2 bolts.

- Tighten to 10 Nm (89 lb-in).

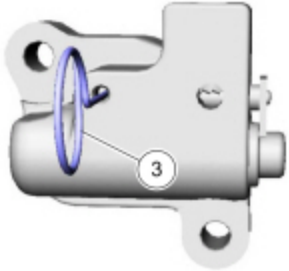
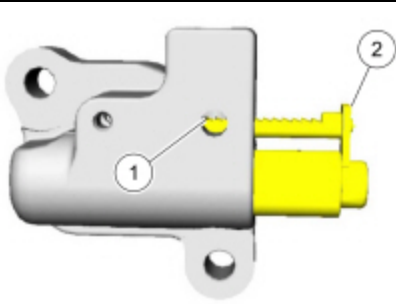


19. Install the primary timing chain tensioner arm.



20. Reset the primary timing chain tensioner.

1. Release the ratchet detent.
2. Using a soft-jawed vise, compress the ratchet plunger.
3. Align the hole in the ratchet plunger with the hole in the tensioner housing and Install a suitable lockpin.

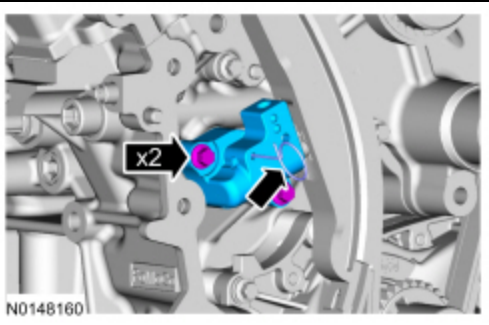


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21. **NOTE:** It may be necessary to rotate the camshafts slightly to remove slack from the timing chain to install the tensioner.

Install the primary timing chain tensioner and the 2 bolts.

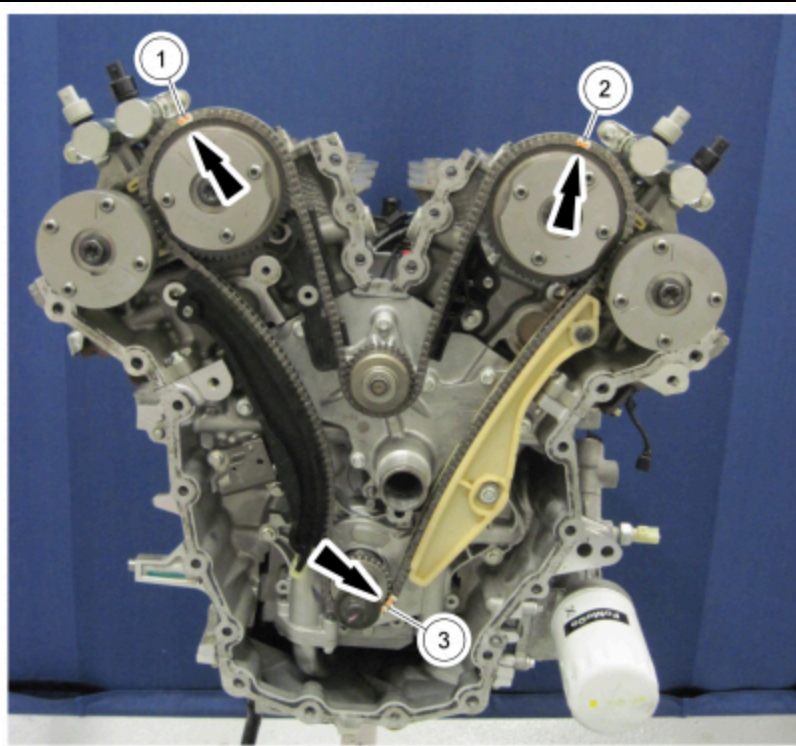
- Tighten to 10 Nm (89 lb-in).
- Remove the lockpin.



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22. As a post-check, verify correct alignment of all timing marks.

- There are 48 links in between the RH intake VCT assembly colored link (1) and the LH intake VCT assembly colored link (2).
- There are 35 links in between LH intake VCT assembly colored link (2) and the 2 crankshaft sprocket links (3).



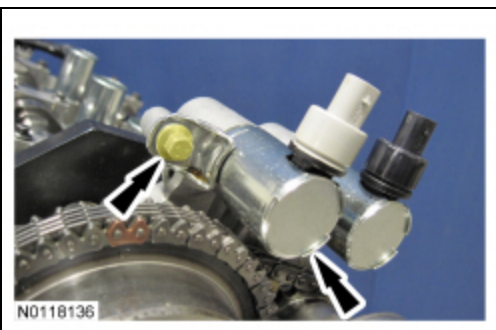
23. **NOTICE:** Do not use excessive force when installing the Variable Camshaft Timing (VCT) oil control solenoid. Damage to the mega cap could cause the cylinder head to be inoperable. If difficult to install the VCT oil control solenoid, inspect the bore and VCT oil control solenoid to ensure there are no burrs, sharp edges or contaminants present on the mating surface. Only clean the external surfaces as necessary.

NOTE: A slight twisting motion will aid in the installation of the VCT oil control solenoid.

NOTE: Keep the VCT oil control solenoid clean of dirt and debris.

Install the LH intake VCT oil control solenoid and the bolt.

- Tighten to 8 Nm (71 lb-in) then an additional 20 degrees.



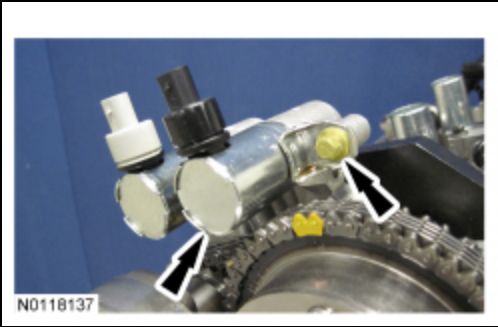
24. **NOTICE:** Do not use excessive force when installing the Variable Camshaft Timing (VCT) oil control solenoid. Damage to the mega cap could cause the cylinder head to be inoperable. If difficult to install the VCT oil control solenoid, inspect the bore and VCT oil control solenoid to ensure there are no burrs, sharp edges or contaminants present on the mating surface. Only clean the external surfaces as necessary.

NOTE: A slight twisting motion will aid in the installation of the VCT oil control solenoid.

NOTE: Keep the VCT oil control solenoid clean of dirt and debris.

Install the RH intake VCT oil control solenoid and the bolt.

- Tighten to 8 Nm (71 lb-in) then an additional 20 degrees.

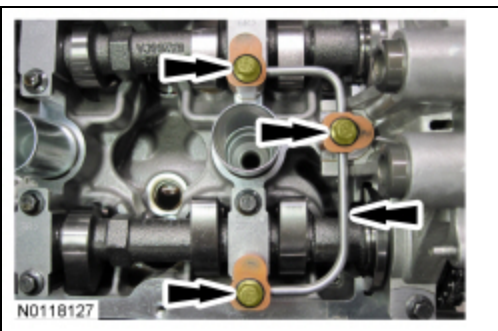


25. Remove the RH Camshaft Holding Tool.

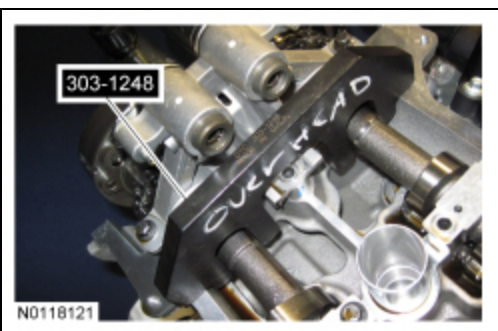


26. Install the RH valve train oil tube and the 3 bolts and tighten in 2 stages.

- Stage 1: Tighten to 8 Nm (71 lb-in).
- Stage 2: Tighten an additional 45 degrees.

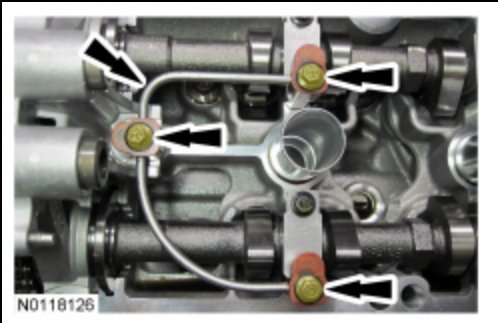


27. Remove the LH Camshaft Holding Tool.



28. Install the LH valve train oil tube and the 3 bolts and tighten in 2 stages.

- Stage 1: Tighten to 8 Nm (71 lb-in).
- Stage 2: Tighten an additional 45 degrees.



29. Install the engine front cover. For additional information, refer to Engine Front Cover in this section.